INCH-POUND

A-A-58090
1 May 97

COMMERCIAL ITEM DESCRIPTION
GENERATOR, SIGNAL, VOR/ILS, AN/ARM-186

The General Services Administration has authorized the use of this Commercial Item Description.

1. SCOPE. This Commercial Item Description (CID) describes a VOR/ILS/MB signal generator. This CID is meant as a minimum requirement for the VOR/ILS/MB signal generator in which only those manufacturers that meet or surpass the following requirements are qualified per this CID. The signal generator shall produce both RF and audio signals for testing VOR, localizer, glide slope, and marker beacon receivers.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and be bid sample tested within the accuracy, limits, and specifications herein.

2.1 Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet this description. The equipment shall be Type II, Class 2, Style A and Color N in accordance with MIL-T-28800.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-T-28800 for the classification as stated herein and as listed below:

2.3 Electromagnetic Interference (EMI). The equipment shall comply with the EMI requirements specified for Class B equipment in MIL-STD-461B, except with the exclusion of test CE06, CE07, RE03, RS03, shall be modified as follows:

a. RE03: Full limits to 10 times the highest used or intentionally generated frequency, or 1 GHz, which ever is greater.

Beneficial comments, recommendation, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: SA-ALC/TILDD. 306 Tinker Drive. Bldg. 207. Kelly AFB, TX 78241-5916 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6625

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b. RS03: Full limits to 1ghz.

2.4 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 115/230 volts AC (+/-10%) at line frequencies of 50Hz to 60Hz. The power input shall be through the rear of the test set. The power input shall be recess mounted on the back of the instrument case. Transient state condition shall meet MIL-T-28800 requirements for the classification specified herein.

2.5 Electrical Power Sources and Connections. Power sources and connections shall be as specified in MIL-T-28800 for Type I, Style A enclosures. All controls and connections shall be on the top of the unit.

2.6 AC Input Power Level. AC input power shall be used to power this Signal Generator and recharge its internal batteries. The equipment shall be capable of operating with an AC power source at nominal 50Hz or 60Hz, single phase with a voltage range of 103.5 to 129 VAC. The equipment shall also operate with an AC power source of: Maximum 440Hz at 122VAC, minimum 360Hz at 104VAC. Automatic recharging of the internal batteries shall occur if the unit is connected to an AC power source, via the AC line cord in both the operational and non-operational states. Operation that would damage the test set or power source at both 115 and 230 volts simultaneously shall not be possible.

2.7 Internal DC Power. DC power shall be supplied by internal, rechargeable storage batteries. The Signal Generator shall be capable of operating on the flight line away from an external AC power source. The rechargeable batteries shall have the capability of operating the unit with full performance for a period of not less than 8 hours over a temperature range of 20°C to 55°C, and a minimum of 2 hours at temperatures of -40°C to -20°C. The battery shall conform to MIL-T-28800 and Requirement 27 of MIL-HDBK-454. The batteries used shall be from a Qualified Parts List and shall have an existing National Stock Number (NSN).

2.8 Shelf Life. The Signal Generator installed batteries shall have a shelf life of at least 1,000 days at 25°C.

2.9 Recharge Time. The charger circuitry shall be such that a condition of charge which shall permit at least 8 hours continuous use of the unit, within all specification requirements, shall be reached in no more than 16 hours following an 8-hour period of use. Under no condition of discharge shall the charging rate applied to the battery exceed the battery manufacturer’s recommended maximum rate, nor shall an indefinite charging period result in any damage to the battery or any other components.

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2.10 Battery Conservation. The Signal Generator shall cycle to the Off mode after 15 minutes of switch inactivity to conserve internal battery power.

2.11 Battery Indication. The Signal Generator shall provide a sunlight readable visual indication on the front panel showing the relative charge state of the internal batteries.

2.12 Power On/Off. The Signal Generator shall provide a sunlight readable visual display of READY on the front panel.

2.13 Battery Power Cutoff Switch. A battery power cutoff switch shall be provided which puts the unit in standby mode when the lid of the combination case is closed.

2.14 Battery Compartment. The battery compartment shall be in accordance with MIL-T-28800 for Type I battery compartment. The battery compartment shall be so designed that battery removal/replacement is easily accomplished by removing one external panel. The battery connector should be of a quick connect (non-solderable) terminal or plug type.

2.15 Battery Charger. An internal battery charger in accordance with MIL-T-28800, Type I, shall be provided with sufficient regulation to prevent overcharging of the batteries specified in 2.7.

2.16 Temperature Requirements.

2.16.1 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment or -40°C to +55°C.

2.16.2 Nonoperating Temperature. The equipment shall meet its performance and accuracy requirements in a nonoperating environment of -50°C to +71°C.

2.17 Mean Time Between Failures (MTBF). The equipment MTBF shall be a minimum of 2,000 hours. The MTBF lower test shall be 2000 hours and the upper test shall be 4,000 hours.

2.18 Pre-conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, the thermal cycling being accomplished in the sample testing phase.

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2.19 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours.

2.19.1 Mean Time To Repair (MTTR). The equipment MTTR shall be as follows: The mean correct maintenance time, H_0 in MIL-STD-471, shall be 45 minutes. The maximum tolerable mean corrective maintenance time, H_1 in MIL-STD 471, shall be 60 minutes. The maximum corrective maintenance time shall not exceed 90 minutes. further requirements and conditions specified in MIL-T-28800 shall apply.

2.19.2 Special Alignment Tools. No special tools shall be required for alignment or adjustment except those furnished with the test set.

2.19.3 Test Points. Test points shall be provided in accordance with the requirements of MIL-T-28800 to facilitate effective maintenance. Test points shall be provided on the overall test set chassis and on each repairable module assembly. The test points shall be conspicuously and precisely marked and shall enable complete maintenance of the simulator be performed at intermediate and depot levels using only general purpose test equipment and precision measuring equipment normally available at these locations.

2.20 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm up period specified in MIL-T-28800 under all combinations of input power conditions, output signal conditions and operating service conditions as specified in MIL-T-28800.

2.20.1 Marker Beacon Generator. A Marker Beacon (MB) generator shall be provided and selected by means of an ON/OFF control or function selector located on the front panel and labeled as such.

2.20.1.1 RF Frequency. The MB generator shall provide continuous sine waves at the fixed frequency of 75.0 MHz ±0.005%. This frequency referred as RF herein shall be modulated as specified herein.

2.20.1.1.1 RF Frequency Stability. The variation in frequency of the RF output frequency shall not exceed ±0.005%.

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2.20.1.1.2 RF Spectral Purity. At the frequency specified in 2.20.1.1, the following specification shall be met:

2.20.1.1.2.1 Harmonics: The level of any harmonic on the desired RF signal shall be at least 30 dB below the output level of the desired fundamental RF signal.

2.20.1.1.2.2 Spurious Signals. The total of all spurious signals, including sub harmonics, accompanying the desired RF signal shall be at least 35 dB below the output level of the desired RF signal at ±15 KHz and at least 40 dB below the carrier output level at ±25.0 KHz in a 300 Hz resolution bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.1.1.2.3 Broadband Noise. The level of any broadband noise on the desired RF signal shall be at least 90 dB below the carrier output level measured in a 1Hz noise bandwidth at more than 50 KHz from the carrier frequency, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.1.2 RF Output For MB, GS, LOC, and VOR.

2.20.1.2.1 RF Output Attenuator. The RF level shall be adjustable in not greater than 1 dB increments from 0 to -110 dBm ±2.0 dBm when measured into a 50 ohm load at the RF IN/OUT connector, or at the output to the antenna, depending on the design.

2.20.1.2.2 RF In/Out Connector/Antennae.

2.20.1.2.2.1 Connector. The RF IN/OUT connectors for the detachable antennae are required by the test set's designs. The connectors shall be a stainless steel chassis mounting type and shall be labeled "RF IN/OUT."

2.20.1.2.2.2 Antennae. Antennae shall be provided and shall be capable of radiating the MB, GS, LOC and VOR frequencies specified in this specification when they are modulated as specified in this specification. The antennae shall be either collapsible into the test set or detachable. The surface-mounted connector or remaining protrusion of the collapsed antenna shall not extend over 0.25 inches from the test set. If the antenna must be longer than 12.0 inches, it shall be telescoping. It shall be mechanically and electrically compatible with the connector specified in 2.20.1.2.2.1.

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2.20.1.2.2.3 RF Output Level. The level of the unmodulated RF signal shall be not less than -20 dBm when measured into a 50 ohm load. The test set shall radiate sufficient RF output power from the antenna, at a 0 dBm attenuation setting, to perform all testing functions while being operated from within the cockpit of the aircraft under test (AUT). The contractor shall submit certification that their test set has this one-man testing capability. No cabling or other physical connections from inside AUT to outside AUT shall be required.

2.20.1.2.2.4 Output Impedance. The output impedance of the RF signal circuit shall be 50 ohms and shall be matched to the antennae.

2.20.1.3, Marker Tones. The MB carrier shall be amplitude modulated by the following audio frequency (AF) tones:

a. 400 Hz sine wave (outer marker beacon).

b. 1300 Hz sine wave (middle marker beacon).

c. 3000 Hz sine wave (inner marker beacon).

2.20.1.3.1 Audio Frequency Accuracy. The audio signals specified in 2.20.1.3 shall be accurate to within ±2.0%.

2.20.1.3.2 Audio Frequency Spectral Purity. For each of the audio signals specified in 2.20.1.3, the total distortion due to harmonics, noise, and spurious signals shall be less than 10.0%.

2.20.1.3.3 Modulation Depth. Each marker tone shall be capable of amplitude modulating the carrier with a modulation level range from 95.0 ±2%.

2.20.1.3.3.1 Envelope Distortion. The distortion of the modulation envelope on the RF carrier shall be less than 10%.

2.20.1.3.4 Tone Selection. Each tone shall be selected by means of control(s) with each control or controls position properly labeled to indicate the marker beacon frequency or position, (i.e. outer, middle, inner).

2.20.2 Glide Slope Generator.

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2.20.2.1 Glide Slope Generator Selection. The Glide Slope (GS) generator shall be selected by means of an ON/OFF control or function selector located on the front panel and labeled as such.

2.20.2.2 RF Frequency Range. The GS generator shall provide continuous sine waves over the frequency range of 334.7 MHz ±0.0025% to 335.55 MHz ±0.0025%. These frequencies, referred to RF herein, shall be modulated as specified herein. These frequencies shall be paired as: 334.7/108.1 MHz and 334.55/108.15 MHz.

2.20.2.2.1 RF Frequency Selection. The output frequency shall be selected by means of a control located on the front panel. The positions of the control shall be labeled with the appropriate frequency pairs.

2.20.2.2.2 RF Frequency Stability. The variation in frequency of the RF output frequency shall not exceed ±0.0025%.

2.20.2.2.3 RF Spectral Purity. At the frequencies specified in 2.20.2.2, the following specifications shall be met:

2.20.2.2.3.1 Harmonics. The level of any harmonic on the desired RF signal shall be at least 25 dB below the output level of the desired fundamental RF signal.

2.20.2.2.3.2 Spurious Signals. The total of all spurious signals, including subharmonics, accompanying the desired RF signal shall be at least 35 dB below the output level of the desired RF signal at ±15 KHz and at least 40 dB below the carrier output level at ±25.0 KHz in a 300 Hz resolution bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.2.2.3.3 Broadband Noise. The level of any broadband noise on the desired RF signal shall be at least 90 dBbelow the carrier output level (measured in a 1 Hz noise bandwidth at less than 50 KHz from the carrier frequency), as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.2.3 Glide Slope Tones. The GS carriers shall be amplitude modulated by 90, 150, and 1020 Hz sine wave audio frequency (AF) tones.

2.20.2.3.1 Audio Frequency Accuracy. The audio signals specified in 2.20.2.3 shall be accurate to within ±0.01%.

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2.20.2.3.2 Audio Frequency Spectral Purity. For each of the audio signals specified in 2.20.2.3, the total distortion due to harmonics, noise, and spurious signals shall be less than 5.0% for the 90 and 150 Hz signals; less than 10.0% for the 1020 Hz signal.

2.20.2.3.3 Modulation Depth. The 90 Hz and 150 Hz signals shall be used to amplitude modulate the GS frequencies specified in 2.20.2.3 and shall be supplied both separately and together. Each signal shall be of sufficient magnitude to modulate the carrier to a preset dept of 40%.

2.20.2.3.4 90 and 150 Hz Phase Error. The phase error of the 90 and 150 Hz signals shall mot exceed 10.0°. Relative phase is defined as the difference in 30 Hz degrees of the points at which the 90 and 150 Hz waveform crosses the zero axis in the same direction at the same time.

2.20.2.3.5 DDM Control. The output ratio of the 90 Hz signals shall have presets at 0.175 ±0.025 DDM, at both 90 Hz predominant and 150 Hz predominant, 0.000 ±0.10 DDM on course ("OC"). For purposes of determining DDM, the base shall be 0.40 for the GS function.

2.20.2.3.5.1 Variable DDM Control. A control shall be provided to continuously vary the DDM from 0.175 ±0.035 DDM, 90 Hz predominant to 0.175 ±0.035 DDM, 150 Hz predominant. This control shall have an off position in which the variable DDM circuitry is not operative.

2.20.2.3.6 90 and 150 Hz Delete. Two front panel controls shall be provided to delete the 90 and 150Hz tones independently.

2.20.3 Localizer Generator.

2.20.3.1 Localizer Generator Selector. The LOC generator shall be selected by means os an ON-OFF control or function selector located on the front panel and labeled as such.

2.20.3.2 RF Frequency Range. The LOC generator shall provide continuous sine waves over the frequency range from 108.10 MHz ±0.0025%. These frequencies, referred to as RF herein, shall be modulated as specified herein. These frequencies shall be paired with the GS frequencies as specified in 2.20.2.3.

2.20.3.2.1 RF Frequency Selection. The output frequency shall be selected as specified in 2.20.3.2.

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2.20.3.2.2 RF Frequency Stability. The variation in frequency of the RF output frequency shall not exceed ±0.0025%.

2.20.3.2.3 RF Spectral Purity. At the frequencies specified in 2.20.3.2 the following specifications shall be met:

2.20.3.2.3.1 Haromonics. The level of any harmonic on the desired RF signal shall be at least 30 dB below the ouptut level of the desired fundamental RF signal.

2.20.3.2.3.2 Spurious Signals. The total of all spurious signals, including subharmonics, accompanying the desired RF signal shall be at least 75 dB below the output level of the desired RF signal at ±25.0 KHz in a 300 Hz resolution bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.3.2.3.3 Broadband Noise. The level of any broadband noise on the desired RF signal shall be at least 90 dB below the carrier output level (measured in a 1Hz noise bandwidth at more than 15 KHz from the carrier frequency), as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.3.3 Localizer Tones. The LOC carriers shall be amplitude modulated by 90, 150, and 1020 Hz sine wave audio frequency (AF) tones.

2.20.3.3.1 Audio Frequency Accuracy. The 90 and 150 Hz signals shall be accurate to within ±0.01%. The 1020 Hz signal shall be accurate to within ±2.0%.

2.20.3.3.2 Audio Frequency Spectral Purity. For each of the audio signals specified in 2.20.3.3, the total distortion due to harmonics, noise and spurious signal shall be less than 5% for the 90 and 150 Hz signals at preset 20% AM, less than 1.0% in the variable range from 5 to 40% AM, and less than 10% for the 1020 Hz signal at preset 30% AM.

2.20.3.3.3 Modulation Depth. The 90 and 150 Hz signals shall be used to amplitude modulate the LOC frequencies specified in 2.20.3.3. The 90 and 150 Hz signals shall be supplied both separately and together and be of suffiecient magnitude to modulate the carrier to a preset depth of 20%. They shall also vary simultaneously and proportionately.

2.20.3.3.4 90 and 150 Hz Phase Error. The phase error of the 90 and 150 Hz signals shall be as specified in 2.20.2.3.4.

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2.20.3.3.5 DDM Control. The output ratio of the 90 and 150 Hz signals shall have at least the following presets 0.155+0.020 DDM, for both 90 Hz dominant and 150 Hz predominant and 0.000 ±0.100 DDM. For purposes of determining DDM, the base shall be 0.20 when the function selector is in the LOC (localizer) function.

2.20.3.3.5.1 Variable DDM Control. A control shall be provided to continuously vary the DDM from 0.155 ±0.030 DDM 90 Hz predominant to 0.155 ±0.020 DDM, 150 Hz predominant. This control shall have an off position in which the variable DDM circuitry is not operative.

2.20.3.3.6 90 and 150 Hz Delete. Front panel controls shall be provided to delete the 90 and 150 Hz tones independently.

2.20.3.3.7 ID Tone. The 1020 Hz signal shall amplitude modulate localizer frequencies, either separately or with the LOC modulation present and shall be selectable by a front panel ON-OFF control or function selector.

2.20.3.3.7.1 Modulation. The depth of modulation of the localizer frequencies by the 1020 Hz signal shall be fixed at a modulation level of 20.0% ±5.0% with less than 5.0% RF tone distortion.

2.20.4 VHF Omnirange Generator.

2.20.4.1 VOR Generator Selection. The VOR signal generator shall be selected by means of an ON-OFF control located on the front panel.

2.20.4.2 RF Frequency Range. The VOR signal generator shall provide continuous sine waves over the frequency range from 108.00 to 108.05 MHz ±0.0025%. These frequencies, referred to as RF herein, shall be modulated as specified herein.

2.20.4.2.1 RF Frequency Selection. The output frequency shall be selected by means of a front panel control.

2.20.4.2.2 RF Frequency Stability. The variation in frequency of the RF output frequency shall not exceed +0.0025%.

2.20.4.2.3 RF Spectral Purity. At the frequencies specified in 2.20.4.2, the following specifications shall be met:

2.20.4.2.3.1 Harmonics. The level of any harmonic on the desired RF signal shall be at least 30 dB below the output level of the desired fundamental RF signal.

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2.20.4.2.3.2 Spurious Signals. The total of all spurious signals, including subharomonics, accompanying the desired RF signal shall be at least 35 dB below the output level of the desired RF signal at ±15KHz and at least 40 dB below the carrier output level at ±25.0 KHz in a 300 Hz resolution Bandwidth, as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.4.2.3.3 Broadband Noise. The level of any broadband noise on the desired RF signal shall be at least 90 dB below the carrier outpur level (measured in a 1 Hz noise bandwidth at more than 15 KHz from the carrier frequency), as measured at the RF IN/OUT connector and into a 50 ohm load.

2.20.4.3 VOR Tones. The VOR carriers shall be amplitude modulated by the following AF tones:

a. 30 Hz sine wave (variable phase).

b. 9960 Hz sine wave (frequency modulated by a 30 Hz reference phase signal).

c. 1020 Hz sine wave.

2.20.4.3.1 Audio Frequency Accuracy. The 30 Hz variable phase, the 30 Hz reference phase, and the 9960 Hz shall be accurate to within ±0.01%, and the 1020 Hz signals shall be accurate to within ±2.0%.

2.20.4.3.2 Audio Frequency Spectral Purity For each of the audio signals specified in 2.20.4.3, the total distortion due to harmonics, noise, and spurious signals shall be less than 5.0% the level of the fundamental audio frequency in a bandwidth of twice the fundamental frequency.

2.20.4.3.2.1 Incidental AM Due to FM. With the 9960 Hz signal frequency modulated with a deviation ratio of 16 ±1 by a 30 Hz signal, the amplitude modulation of the 9960 Hz signal shall not exceed 1.0%.

2.20.4.3.3 9960 Hz FM Subcarrier. The 9960 Hz subcarrier shall be frequency modulated by a 30 Hz fixed reference phase signal, with a peak deviation 480 ±30 Hz. The 9960 Hz FM signal shall amplitude modulate the RF carriers.

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2.20.4.3.4 30 Hz Variable Phase Signal. A second 30 Hz signal shall be provided which is variable in phase with respect to the 30 Hz fixed phase reference signal from 0° to 315.00° lagging in ±45.0° steps.

2.20.4.3.5 30 Hz Frequency Equality. The 30 Hz reference signal and 30 Hz variable signal shall be coherent.

2.20.4.3.6 Bearing Signal. The 9960 Hz subcarrier (9960 FM signal), which is frequency modulated by the 30 Hz reference signal, and the 30 Hz variable phase signal shall amplitude the RF carriers simultaneously and proportionately.

2.20.4.3.6.1 Modulation Depth. The 9960 Hz FM signal and the 30 Hz variable signal shall each be of sufficient magnitude to amplitude modulate the RF carriers to the depth of preset 30%.

2.20.4.3.6.2 Bearing Selector. A control shall be provided to vary the bearing from 000.0 to 315.0° in ±45° steps.

2.20.4.3.6.2.1 Bearing Indicator. The bearing indicator shall have a resolution of 45° or better and indicate directly in degrees.

2.20.4.3.6.2.2 Bearing Accuracy. The bearing of the VOR signal on the RF carrier shall be within ±0.33° of the audio bearing as set by a VOR standard (Collins Radio Group Model 478A/2 or equal).

2.20.4.3.6.2.3 Bearing Stability. The VOR bearing shall not vary more than ±1.0° under all combinations of environmental conditions.

2.20.4.3.6.2.4 Variable Bearing Control. A control shall be provided to continuously vary the VOR bearing 10° ±2° either clockwise or counterclockwise about the selected bearing. This control shall have an off position in which the variable bearing circuitry is not operative.

2.20.4.3.7 30 Hz Variable and 30 Hz Reference Delete. Front panel control shall be provided to delete the 30 Hz variable and 30 Hz reference signals independently and simultaneously.

2.20.4.3.8 ID Tone. The 1020 Hz shall amplitude modulate VOR frequencies both separately and with the VOR modulation present. An ON-OFF front panel control shall be provided for this signal.

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2.20.4.3.8.1 Modulation. The depth of modulation of the VOR frequencies fy the 1020 Hz signal shall be fixed at a modulation level of 20.0% ±5.0% with less than 5% RF distortion.

2.20.5 Interface. The VOR/ILS Signal Generator shall provide a IEEE-488 interface function and connector.

2.21 Dimensions and Weight. The size and weight of the signal generator shall be consistent with current commercial capabilities, refer to paragraph 3.21.1 and 3.21.2 below.

2.21.1 Equipment Dimensions. The maximum height of the signal generator should be approximately 8 inches, with a nominal volume of 1500 cubic inches.

2.21.2 Equipment Weight. The weight including all accessories, antennae, batteries, and battery charger (excluding manuals), shall not exceed 22 lbs.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 The contracting officer has the option of accepting or rejecting the product.

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5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of Documents.

6.1.1 Military Specifications and Standards and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094

6.1.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.1.3 ANSI Standards are available from the American National Standards Institute, 1430 Broadway, New York, NY 10018.

6.2 Known Acceptable Products. The following products are known to have previously met the requirements this CID and need not resubmit bid sample or certification. All other manufacturers must meet the requirements of this CID and submit to bid sample testing and certification as required. The list may not be indicative of all possible NSNs associated with the CID.

Manufacturer	Model	NSN
IFR	NAV 401L	6625-01-009-2906
TEL Instrument	T-30CM Mode 2	6625-01-421-5845
Procurement Document	A-A-58090	6625-01-037-5344

6.3 Part Identification Number (PIN). National Stock Numbers of products meeting the requirements of this CID will be assigned part number AA58090 in the Federal Cataloging System. This action is for government purposes and does not constitute a requirement for the contractor.

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MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - MI

Review Activities:
OSD/SO

Civil Agency Coordinating Activity:
GSA - FSS

Preparing Activity:
Air Force - 82

Project No. 6625-F066

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** A-A-58090	**2. DOCUMENT DATE (YYMMDD)** 970501

3. DOCUMENT TITLE GENERATOR, SIGNAL, VOR/ILS, AN/ARM-186

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7. DATE SUBMITTED** (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)* SA-ALC/TILDD 306 TINKER DRIVE BLDG 207 KELLY AFB, TX 78241-5916	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** DEFENSE QUALITY AND STANDARDIZATION OFFICE 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22401-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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